# DNS Behaviour Monitor

# Analysing Normal and Unusual DNS Patterns in Network Traffic

By

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# Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

**SAJAN TAMANG**

**Date:** 29th of January

# Acknowledgements

# Abstract

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# 1. Introduction

## 1.1 Introduction

The Domain Name System (DNS) is a fundamental component of Internet communication, enabling users and applications to access networked resources by translating human-readable domain names into numerical Internet Protocol (IP) addresses. Every interaction with web services, cloud platforms, and many background system processes depends on DNS resolution as an initial step. As a result, DNS queries are generated continuously by operating systems, applications, and automated services, producing a detailed and persistent stream of network metadata.

Despite its central role, DNS is typically treated as a transparent infrastructure service that is only examined when failures occur or when security incidents are suspected. In everyday networks—particularly small-scale, educational, or laboratory environments—DNS traffic is rarely analysed beyond basic operational metrics such as total query counts or blocked domains. This limited examination reduces opportunities to understand how systems normally interact with DNS and how changes in user behaviour, application design, or configuration influence DNS usage patterns.

This project positions DNS traffic as a behavioural data source rather than a purely operational or security artefact. By analysing DNS logs collected within a controlled environment, the project seeks to characterise normal DNS behaviour and highlight unusual patterns without assuming malicious intent. The emphasis is on understanding variation and deviation in DNS usage as an analytical and educational exercise, rather than developing a detection or enforcement system.

## 1.2 Problem Statement and Research Question

DNS traffic generally follows predictable behavioural patterns influenced by factors such as user activity, application behaviour, caching mechanisms, and resolver configuration. However, deviations from these patterns may arise due to benign causes including software updates, automation, experimental applications, or misconfigurations. In many cases, such deviations are either ignored or immediately framed through a security-focused lens.

Existing DNS monitoring tools and research outputs primarily target enterprise-scale environments or security detection use cases. These systems often prioritise alerting, classification, or threat identification, and typically assume access to large datasets and specialist infrastructure. As a result, they are poorly suited to small or controlled environments where the primary goal is exploratory understanding rather than detection or enforcement.

The problem addressed by this project is the lack of lightweight, transparent approaches for analysing DNS behaviour in controlled environments that focus on behavioural understanding rather than security classification. There is a need for a system that enables users to observe, visualise, and interpret DNS behaviour patterns,

and to identify unusual activity relative to established baselines without attributing intent.

**Research Question:**

How can DNS behaviour be characterised and visualised in a controlled environment in order to distinguish normal usage patterns from unusual DNS activity without relying on security-focused detection techniques?

# 1.3 Aims and Objectives

## 1.3.1 Aims

The overarching aim of this project is to improve understanding of Domain Name System behaviour by analysing DNS traffic patterns within a controlled network environment. Rather than focusing on security detection or threat classification, the project adopts an exploratory and analytical perspective to examine how DNS is used during normal system operation and how deviations from typical behaviour may arise.

Specifically, the aims of this project are to:

•Analyse DNS traffic to characterise normal behavioural patterns, including temporal, structural, and distributional characteristics.

•Identify and highlight unusual DNS behaviours by comparing observed activity against established behavioural baselines, without assigning malicious intent.

•Design and implement a lightweight DNS behaviour monitoring tool that transforms raw DNS logs into interpretable visual representations to support behavioural insight and learning.

## 1.3.2 Objectives

To achieve the stated aims, the project is structured around the following objectives:

•Collect DNS query logs from a controlled local environment using lightweight DNS logging tools such as Pi-hole or dnsmasq.

•Pre-process and normalise raw DNS logs to enable consistent analysis, including extraction of timestamps, queried domains, and DNS response codes.

•Establish baseline DNS behaviour metrics, including query volume over time, response code distributions, top-level domain (TLD) usage, and domain name structural characteristics.

•Define transparent heuristic criteria for identifying unusual DNS behaviour, such as repeated failed lookups, excessive domain name length, uncommon TLD usage, or abnormal query bursts.

•Develop a Python-based DNS behaviour analysis engine capable of computing behavioural metrics and highlighting deviations relative to established baselines.

•Design and implement a lightweight dashboard to visualise DNS behaviour trends and highlighted observations in an interpretable manner.

•Plan and conduct an evaluation of the system using multiple DNS datasets generated under varied but controlled conditions.

### 1.3.3 Scope Clarification

This project explicitly focuses on DNS behaviour analysis rather than security detection or threat classification. While unusual DNS patterns may resemble indicators discussed in security literature, the system does not attempt to label activity as malicious. All observations are presented as behavioural deviations that require contextual interpretation. The scope is deliberately limited to controlled environments in order to support transparency, reproducibility, and ethical data handling.

# 2. Background and Context

## 2.1 Background

The Domain Name System (DNS) is a fundamental Internet service that enables users and applications to access networked resources by translating human-readable domain names into numerical Internet Protocol (IP) addresses. Every interaction with web services, cloud platforms, and many background system processes relies on DNS resolution as an initial step. Consequently, DNS queries are generated continuously by operating systems, applications, and automated services, producing a rich and persistent source of network behaviour data.

Despite its central role, DNS is commonly treated as a transparent infrastructure component that is only examined when failures occur or when security incidents are suspected. In many environments—particularly small networks, educational settings, and local laboratory deployments—DNS traffic is rarely analysed beyond basic operational metrics such as total query counts or blocked domains. This limited examination restricts understanding of how systems normally interact with DNS and how changes in configuration, application behaviour, or user activity influence DNS usage patterns.

DNS behaviour is shaped by several interacting factors, including resolver design, caching mechanisms, application polling behaviour, and background system processes. Caching, in particular, introduces natural variability into DNS traffic, causing periodic bursts and repeated queries when cache entries expire. As a result, deviations from expected DNS patterns are not inherently malicious and may arise from benign operational causes such as software updates, service discovery mechanisms, or configuration errors. Understanding these baseline behaviours is therefore essential before any meaningful interpretation of unusual DNS activity can be attempted.

This project adopts a behavioural perspective on DNS analysis by treating DNS logs as an empirical data source rather than as indicators of faults or threats. By analysing DNS logs collected within a controlled environment, the project aims to characterise normal DNS behaviour patterns and to highlight deviations without attributing intent. This contextual and exploratory approach underpins the analytical methods and system design choices described in later sections.

# 2.2 Literature Review

## Foundational DNS Architecture and Behaviour

**P. V. Mockapetris, "Domain Names – Concepts and Facilities," RFC 1034, 1987 [1].**

Mockapetris introduced the Domain Name System as a distributed and hierarchical naming architecture designed to provide scalability, resilience, and decentralised administration. The RFC explains how delegation, recursion, and caching fundamentally shape DNS behaviour and directly influence observable DNS traffic patterns. Although the document does not address monitoring or behavioural analysis, it establishes the conceptual baseline for defining what constitutes normal DNS behaviour. A key limitation is that the RFC is specification-oriented and does not consider empirical traffic analysis or behavioural variation. This project builds upon these architectural principles by translating DNS concepts into measurable behavioural metrics, enabling DNS logs to be analysed as behavioural data rather than purely protocol events.

**P. V. Mockapetris, "Domain Names – Implementation and Specification," RFC 1035, 1987 [2].**

RFC 1035 defines DNS message formats, response codes, and resolver behaviour, making it essential for interpreting DNS logs accurately. Response codes such as NOERROR, NXDOMAIN, and SERVFAIL are specified at the protocol level and frequently appear in DNS traffic datasets. While the RFC provides precise technical definitions, it does not offer guidance on how these responses should be interpreted over time or across datasets. This limitation is addressed in the present project by analysing response codes statistically, establishing behavioural baselines, and highlighting deviations rather than treating individual responses in isolation.

## DNS Performance and Traffic Characteristics

**J. Jung, E. Sit, H. Balakrishnan, and R. Morris, "DNS Performance and the Effectiveness of Caching," IEEE/ACM Transactions on Networking, 2002 [3].**

This study presents one of the earliest large-scale empirical analyses of DNS traffic and demonstrates the significant impact of caching on query frequency, latency, and burst behaviour. The authors show that repeated queries and sudden spikes are often legitimate outcomes of cache expiry rather than abnormal activity. While the study focuses on performance optimisation, it provides important insight into why DNS traffic naturally exhibits variability and burstiness. Its limitation lies in the lack of behavioural interpretation or visual analysis. This project extends these findings by incorporating

time-based visualisation and baseline comparison to contextualise caching-related behaviour in small-scale environments.

**M. Crovella and B. Krishnamurthy, *Internet Measurement: Infrastructure, Traffic and Applications*, Wiley, 2006 [4].**

Crovella and Krishnamurthy provide a comprehensive overview of Internet traffic measurement methodologies, including baseline construction, heavy-tailed distributions, and observational bias. Although not specific to DNS, the book establishes methodological principles that are directly applicable to DNS behaviour analysis, particularly the need to avoid overinterpreting anomalies. A limitation is the absence of DNS-specific heuristics or examples. This project applies these general measurement principles directly to DNS logs, ensuring that unusual behaviour is interpreted relative to established baselines rather than treated as definitive indicators.

## Large-Scale DNS Behaviour and Measurement

**DNS Operations, Analysis, and Research Center (DNS-OARC), "DNS Behaviour Analysis Reports," 2023 [5].**

DNS-OARC publishes authoritative DNS measurement reports based on large-scale datasets collected from operational networks worldwide. These reports highlight global DNS behaviour patterns, error rates, and operational anomalies, demonstrating that DNS traffic exhibits consistent baseline characteristics even at Internet scale. However, DNS-OARC's analyses are primarily intended for professional operators and researchers and assume access to large datasets and specialist expertise. This project draws conceptual inspiration from DNS-OARC's baseline-driven approach while adapting it to a lightweight, transparent system suitable for controlled and educational environments.

**Cloudflare Research, "Understanding DNS Traffic Patterns," 2023 [6].**

Cloudflare's research and public reporting provide insight into DNS behaviour at massive scale, illustrating temporal trends, geographic variation, and the dominance of specific domain categories. These studies reinforce the idea that DNS behaviour is dynamic and influenced by external factors such as application design and infrastructure changes. However, Cloudflare's analytics lack methodological transparency and are not designed for small-scale analysis. This project differs by prioritising interpretability and reproducibility, applying similar behavioural concepts within a controlled environment where analytical assumptions can be explicitly documented.

## Practical DNS Logging and Tooling

**Pi-hole Developers, "Pi-hole Documentation," 2024 [7].**

Pi-hole is a widely used DNS logging and filtering platform that provides detailed query records and basic statistical summaries. Its documentation demonstrates how DNS

logs can be efficiently collected in small networks, making it a practical data source for this project. However, Pi-hole's analytics are limited to predefined operational metrics and do not support behavioural baseline modelling or contextual interpretation of unusual patterns. This project extends Pi-hole's functionality by treating its logs as raw behavioural data and applying custom analysis to identify trends and deviations in DNS behaviour rather than focusing on filtering outcomes.

## Summary and Implications for This Project

The reviewed literature establishes a strong foundation in DNS architecture, performance characteristics, and large-scale measurement. However, existing research and tools largely focus on protocol specification, performance optimisation, or Internet-scale analytics. There is limited emphasis on exploratory DNS behaviour analysis in small or controlled environments using lightweight and interpretable methods.

This project addresses this gap by integrating foundational DNS concepts with empirical log analysis and visualisation. The literature directly informs the project's methodological choices, including baseline-driven analysis, heuristic interpretation of unusual behaviour, and the design of a lightweight dashboard focused on behavioural insight rather than detection or enforcement.

## Critical Discussion and Synthesis of the Literature

Across the reviewed literature, DNS is consistently treated as a critical infrastructure component, yet its behavioural characteristics are often secondary to concerns of correctness, performance, or security. Foundational RFCs define how DNS should operate but do not address how DNS behaviour varies over time or how it should be analysed empirically. Performance-focused studies provide valuable insight into caching effects and traffic variability but frame behaviour primarily through the lens of optimisation rather than understanding.

Large-scale measurement initiatives conducted by organisations such as DNS-OARC and Cloudflare demonstrate that DNS behaviour is measurable and that baseline patterns exist even at Internet scale. However, these approaches assume access to extensive datasets and specialist infrastructure, limiting their applicability to small networks. Practical tools such as Pi-hole provide accessible logging but lack analytical depth and contextual interpretation.

A key limitation across much of the literature is the tendency to interpret deviations through a security or fault-detection perspective. This project adopts a different stance by treating unusual DNS behaviour as an observation requiring contextual explanation rather than immediate classification. This shift is directly motivated by the literature, which demonstrates that DNS behaviour is influenced by multiple benign factors including caching, application design, and background system processes.

## Implications for Methodology and System Design

The findings of the literature review directly guide the methodological choices and system design adopted in this project. Foundational DNS specifications inform how DNS

logs are parsed and interpreted, ensuring that observed behaviour is aligned with protocol-defined semantics. Performance and measurement studies justify the use of baseline-driven analysis and time-based aggregation rather than static thresholds.

Large-scale DNS studies support the use of distributional metrics such as domain popularity, response code frequency, and temporal patterns, while their limitations motivate the project's focus on lightweight analysis suitable for controlled environments. Practical tools such as Pi-hole demonstrate the feasibility of DNS log collection but highlight the need for custom analysis to extract behavioural insight.

As a result, the project adopts an exploratory data analysis approach, focusing on baseline characterisation, heuristic identification of unusual behaviour, and transparent visualisation. These decisions are grounded in the literature and align directly with the project's aims and objectives.

### Conclusion of the Literature Review

This literature review demonstrates that while DNS has been extensively studied from architectural, performance, and operational perspectives, there is limited emphasis on exploratory DNS behaviour analysis in small or controlled environments. Existing tools and studies either operate at Internet scale or focus on detection and enforcement rather than interpretability.

The reviewed literature highlights the importance of baseline-driven analysis, contextual interpretation, and cautious evaluation of deviations. These insights directly inform the design of the proposed DNS behaviour monitoring tool, which aims to provide transparent, lightweight analysis and visualisation without relying on security-focused assumptions. By addressing the identified gap, this project contributes a novel educational and analytical perspective to DNS behaviour analysis.

# 3. Competitor Analysis / Related Work

## 3.1 Existing DNS Monitoring Tools and Platforms

### 3.1.1 Pi-hole

**How it works:**
Pi-hole functions as a DNS sinkhole that intercepts DNS queries and blocks requests to known advertising and tracking domains. All DNS queries passing through the resolver are logged and summarised through a web-based dashboard.

**Availability and cost:**
Pi-hole is open-source and free to use. It is widely adopted in home networks, laboratories, and educational environments due to its ease of deployment and minimal hardware requirements.

**Strengths:**

• Simple installation and configuration
• Clear operational visibility into DNS queries
• Provides basic metrics such as total queries, top domains, and blocked requests

**Limitations:**

• Analytics are limited to predefined operational summaries
• No baseline modelling or longitudinal comparison
• No contextual interpretation of unusual behaviour
• Primary focus on filtering effectiveness rather than behaviour understanding

**Relevance to this project:**
Pi-hole is used in this project as a DNS log collection mechanism rather than as an analytical solution. The proposed system extends Pi-hole's functionality by transforming its logs into behavioural metrics, establishing baselines, and highlighting deviations with explanatory context.

### 3.1.2 dnsmasq

**How it works:**
dnsmasq is a lightweight DNS forwarder and DHCP server commonly deployed on routers and embedded systems. It produces detailed DNS logs but does not provide built-in analytics or visualisation.

**Availability and cost:**
dnsmasq is open-source and free, making it suitable for small networks and controlled environments.

**Strengths:**

• Lightweight and highly configurable
• Produces detailed DNS logs suitable for analysis
• Widely supported across platforms

**Limitations:**

• No built-in dashboard or analytics
• Requires external tools for interpretation
• High analytical burden on the user

**Relevance to this project:**
dnsmasq provides an alternative DNS log source for the proposed system. The project's analysis engine abstracts log format differences and converts raw dnsmasq logs into structured behavioural insights.

### 3.1.3 Cloudflare DNS Analytics

**How it works:**
Cloudflare DNS Analytics provides insight into DNS traffic processed through Cloudflare's global resolver infrastructure. It visualises metrics such as query volume, latency, and error rates at Internet scale.

**Availability and cost:**
Access to detailed analytics is tied to Cloudflare's managed services and enterprise plans.

**Strengths:**

• High-quality visualisation
• Massive datasets and global perspective
• Strong performance and resilience metrics

**Limitations:**

• Tightly coupled to proprietary infrastructure
• Limited methodological transparency
• Unsuitable for small or offline environments
• Focus on performance and security rather than behavioural exploration

**Relevance to this project:**
Cloudflare's work demonstrates the value of baseline-driven DNS analysis but is not transferable to controlled environments. The proposed system adopts similar behavioural concepts while prioritising transparency, reproducibility, and small-scale applicability.

### 3.1.4 DNS Operations, Analysis, and Research Center (DNS-OARC)

**How it works:**
DNS-OARC operates as a research consortium that produces DNS measurement datasets, tools, and reports to support DNS reliability and operational research at Internet scale.

**Availability and cost:**
Some datasets and tools require paid membership. Outputs are primarily aimed at professional DNS operators and researchers.

**Strengths:**

• Highly authoritative research outputs
• Strong methodological rigour
• Internet-scale behavioural insight

**Limitations:**

• Requires specialist knowledge and infrastructure
• Not interactive or exploratory
• Unsuitable for educational or small-scale deployment

**Relevance to this project:**
DNS-OARC provides conceptual inspiration for baseline-driven analysis. The proposed system adapts these ideas into a lightweight, hands-on tool suitable for controlled environments and learning contexts.

## 3.2 Comparative Feature Analysis

The table below summarises key differences between existing tools and the proposed DNS behaviour monitoring system.

| Feature | Pi-hole | dns-masq | Cloudflare DNS | DNS-OARC | Proposed System |
|---|---|---|---|---|---|
| Open-source | Yes | Yes | No | Partial | Yes |
| Cost | Free | Free | Paid | Membership | Free |
| Behavioural baselines | No | No | Limited | Yes | Yes |
| Unusual behaviour highlighting | No | No | Limited | Yes | Yes |
| Visualisation | Basic | None | Advanced | Limited | Interpretable |
| Scale | Small | Small | Enterprise | Internet | Small / Controlled |
| Transparency | Medium | High | Low | Medium | High |
| Educational suitability | Medium | Low | Low | Low | High |

## 3.3 Identified Gap and Project Contribution

The comparison demonstrates that existing DNS monitoring solutions prioritise either operational filtering, raw logging, enterprise-scale analytics, or research-grade measurement. None are explicitly designed to support exploratory DNS behaviour analysis in controlled or educational environments using lightweight and transparent methods.

The proposed DNS behaviour monitoring system fills this gap by combining accessible DNS log sources with baseline-driven analysis and interpretable visualisation. By focusing on behavioural understanding rather than detection or enforcement, the system complements existing tools and provides a novel analytical perspective aligned with the project's aims and objectives.

# 4. Tools and Methods

## 4.1 Rationale for Technical Tools Selected

### 4.1.1 DNS Log Collection Tools

DNS data for this project is collected using lightweight DNS logging tools, specifically Pi-hole and dnsmasq, deployed within a controlled local environment.

**How DNS data is collected:**
DNS queries generated by normal system usage are logged by the local DNS resolver. The logs capture DNS metadata including timestamps, queried domain names, and DNS response codes. No payload data or content is recorded.

**Rationale for selection:**

• Lightweight and easy to deploy in small networks
• Open-source and freely available
• Well-documented and widely used
• Suitable for reproducible experimentation in controlled environments

**Alternatives considered:**

Enterprise DNS platforms and managed cloud resolvers were considered but rejected due to cost, complexity, and lack of transparency. Such platforms are unsuitable for exploratory behavioural analysis in small-scale or educational contexts.

### 4.1.2 Programming Language and Data Processing

The DNS behaviour analysis engine is implemented using Python.

**Rationale for selection:**

• Strong support for data parsing and manipulation
• Extensive ecosystem of data analysis libraries
• High readability and maintainability
• Widely used in networking and Internet measurement research

**Alternatives considered:**

Lower-level languages such as C or C++ were considered but rejected due to increased development complexity and reduced flexibility for exploratory analysis. Java was also considered but offered no clear advantages for the project scope.

### 4.1.3 Visualisation and Dashboard Framework

A lightweight web-based dashboard is planned using Flask or Streamlit.

**Purpose of the dashboard:**

• Visualise DNS behaviour trends over time
• Present baseline metrics and distributions
• Highlight unusual DNS behaviour with contextual explanations

**Rationale for selection:**

• Rapid development of data-driven interfaces
• Minimal front-end complexity
• Seamless integration with Python-based analysis

**Alternatives considered:**

Full-scale front-end frameworks such as React or Angular were rejected due to unnecessary complexity relative to the project's analytical goals.

### 4.1.4 Data Storage Approach

DNS logs and derived metrics are stored using simple, structured formats such as CSV files or a lightweight local database.

**Rationale for selection:**

• Low data volume in controlled environments
• Supports reproducibility and portability
• Aligns with data minimisation principles

Cloud-based storage solutions were rejected due to privacy considerations and lack of necessity.

## 4.2 Rationale for Research Methods Used

### 4.2.1 Exploratory Data Analysis (EDA)

The project adopts Exploratory Data Analysis (EDA) as its primary research methodology.

EDA is a data analysis approach focused on discovering patterns, trends, and relationships without predefined assumptions. It is particularly suitable when system behaviour is influenced by multiple interacting factors and is not fully understood in advance.

**Suitability for this project:**

• DNS behaviour varies with context, application design, and configuration
• Avoids assumptions of malicious intent
• Supports behavioural understanding rather than classification

### 4.2.2 Baseline-Driven Behaviour Analysis

A baseline-driven approach is used to characterise normal DNS behaviour before identifying deviations.

**Baseline metrics include:**

• DNS query volume over time
• Distribution of DNS response codes
• Top-level domain (TLD) usage
• Domain name length and structural patterns

**Rationale for selection:**

DNS traffic exhibits natural variability due to caching and background processes. Baselines provide essential context for interpreting deviations and reduce the risk of overinterpreting normal variability as abnormal behaviour.

### 4.2.3 Heuristic Identification of Unusual Behaviour

Unusual DNS behaviour is identified using transparent, rule-based heuristics.

**Examples of heuristics used:**

• Repeated NXDOMAIN responses exceeding baseline levels
• Unusually long or deeply nested domain names
• Sudden bursts in DNS query volume
• Rare or uncommon TLD usage

**Alternatives considered:**

Machine learning and anomaly detection techniques were considered but rejected due to the lack of labelled training data, reduced explainability, and misalignment with the project’s educational and analytical goals.

# 4.3 Project Management and Product Lifecycle

The project follows an iterative development lifecycle.

**Project management approach:**

• Incremental implementation of features
• Early proof-of-concept development
• Continuous refinement based on findings and feedback

**Tools used:**

• GitHub for version control
• Lightweight milestone tracking aligned with the project plan

This approach supports early risk identification and ensures that core analytical functionality is validated before full dashboard implementation.

# 4.4 Legal, Ethical and Professional Issues

## 4.4.1 Legal Considerations

### 4.4.1.1 Data Protection and Privacy

The project analyses DNS metadata only. The data collected is limited to:

• Queried domain names
• Timestamps of DNS requests
• DNS response codes
• Aggregate query counts

No payload data, content data, user identifiers, or personally identifiable information (PII) are collected or processed. All DNS logs are generated within a controlled local environment that is owned and administered by the researcher.

Legal compliance measures include:

• Data minimisation through collection of metadata only
• Local storage of DNS logs
• Limited data retention for academic use only
• Anonymisation where applicable

These measures align with data protection principles such as purpose limitation, data minimisation, and storage limitation under relevant data protection regulations.

#### 4.4.1.2 Compliance with Computer Misuse Legislation

The project does not involve unauthorised access to computer systems or networks. Specifically, it does not include:

• Network scanning
• Interception of third-party communications
• Monitoring of external or public networks
• Exploitation of vulnerabilities

All testing is conducted exclusively on systems and networks under the direct control of the researcher. As a result, the project complies with computer misuse legislation and avoids unauthorised monitoring or access.

## 4.4.2 Ethical Considerations

#### 4.4.2.1 Ethical Use of Network Data

DNS metadata can indirectly reveal patterns of system or user behaviour if misused. To mitigate this risk, the project adopts the following ethical safeguards:

• Analysis is performed at an aggregate behavioural level
• No attempt is made to profile individual users or devices
• Results are presented as trends and observations rather than individual activity logs

The project deliberately avoids interpreting unusual DNS behaviour as malicious, reducing the risk of misrepresentation or misuse of analytical outputs.

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• Analysis is performed at an aggregate behavioural level
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• Results are presented as trends and observations rather than individual activity logs

The project deliberately avoids interpreting unusual DNS behaviour as malicious, reducing the risk of misrepresentation or misuse of analytical outputs.

#### 4.4.2.2 Ethical Approval Process

Prior to data collection and evaluation activities, an ethics review was conducted in accordance with University of Roehampton guidelines for undergraduate and post-graduate computing projects. The ethics approval process included:

- Completion and submission of an ethics application form
- Clear definition of DNS data collection scope and analytical methods
- Assessment of privacy, misuse, and participant-related risks
- Approval for data collection within a controlled environment and for planned low-risk usability evaluation

The approved ethics documentation is included in the project appendix, alongside other supporting artefacts referenced throughout the report**.**

#### 4.4.2.3 Informed Consent and Transparency

Two distinct forms of activity are considered within the project: DNS data analysis and planned usability evaluation.

For DNS data analysis, all testing is conducted within a controlled environment managed by the researcher. No external participants are involved, and no personal data belonging to others is collected. Consequently, informed consent is not required for DNS data collection.

For the planned dashboard usability evaluation, voluntary participants (e.g., cyber security students) will be recruited to assess the clarity and interpretability of pre-generated dashboard outputs. Participants will not provide or interact with their own DNS data. Informed consent will be obtained prior to participation, and participation will be voluntary, anonymous, and free from coercion. Participants will be informed of the purpose of the study, their right to withdraw at any time, and how their feedback will be used.

## 4.4.3 Professional Considerations

#### 4.4.3.1 Responsible Use of Tools and Techniques

All tools used in the project are open-source and are employed in accordance with their intended purpose. Proper attribution is provided for all third-party tools, libraries, and frameworks used.

Professional standards are maintained by:

• Clearly documenting assumptions and limitations
• Avoiding exaggerated claims about detection or security capability
• Ensuring transparency in analytical methods and outputs

#### 4.4.3.1 Scope Management and Professional Integrity

The project deliberately avoids positioning itself as a security detection or threat identification system. This decision reflects professional responsibility and ensures that the artefact is not misrepresented or misused beyond its intended analytical and educational purpose.

Unusual DNS behaviour is presented as an observation requiring contextual explanation rather than as evidence of malicious activity.

# 6. Risk Register

## 6.1 Risk Identification and Mitigation Strategy

The primary risks identified for this project fall into the following categories:

• Technical feasibility and implementation complexity
• Data availability and data quality
• Risk of scope creep towards security detection
• Legal and ethical compliance
• Time management and project delivery

To mitigate these risks, an early-stage PoC was developed to validate DNS log parsing, behavioural metric extraction, and heuristic-based observation. This reduced uncertainty and informed refinements to both the project scope and design.

## 6.2 Risk Register Table

| Risk ID | Risk Description | Like-li-hood | Im-pact | Mitigation Strategy | Cur-rent Status |
|---|---|---|---|---|---|
| R1 | DNS logs may not contain sufficient data to identify meaningful behavioural patterns | Medium | High | Early proof-of-concept developed to validate availability of key metrics such as query volume, TLD distribution, and response codes | Mitigated |
| R2 | Selected tools may be unsuitable or overly complex for small-scale environments | Low | Medium | Lightweight, open-source tools (Pi-hole, dnsmasq, Python) selected and validated through early testing | Mitigated |
| R3 | Difficulty interpreting DNS behaviour without misclassifying benign activity | Medium | Medium | Adoption of baseline-driven and heuristic-based analysis rather than security detection | Ongoing |
| R4 | Project scope drifting towards security detection or threat classification | Medium | High | Clear scope definition reinforced throughout aims, objectives, and design decisions | Mitigated |
| R5 | Legal or ethical issues arising from DNS data collection | Low | High | Data collection limited to metadata in a controlled environment; ethics approval obtained | Mitigated |
| R6 | Insufficient time to complete dashboard development | Medium | Medium | Iterative development approach with core analysis prioritised before visualisation | Ongoing |
| R7 | Limited reproducibility across different DNS log formats | Medium | Low | Log parsing designed to be adaptable and tested with multiple log sources | Ongoing |

## 6.3 Role of the Proof-of-Concept in Risk Reduction

The early development of a functional proof-of-concept played a critical role in reducing project risk. The PoC successfully demonstrated that:

•DNS logs generated by Pi-hole and dnsmasq are suitable for behavioural analysis
•Key behavioural metrics can be reliably extracted from lightweight DNS logs
•Heuristic rules can highlight potentially unusual DNS behaviour without complex models

This early validation reduced the likelihood of late-stage technical failure and provided confidence in the feasibility of the proposed system.

# 7. Requirements / Specifications

## 7.1 Intended Users and Personas

Personas are used to represent the intended users of the system and to ensure that design decisions remain user-focused.

### Persona 1: Cyber Security Student

**Background:** Undergraduate cyber security student studying networking concepts
**Technical skill level:** Intermediate

**Goals:**
• Understand how DNS behaves under normal conditions
• Observe how DNS traffic patterns change over time
• Learn how unusual DNS behaviour emerges without assuming malicious intent

**Frustrations:**
•Existing tools focus on blocking or detection rather than explanation
• Enterprise tools are complex and opaque

**How the system helps:**
• Provides clear visualisations of DNS behaviour
• Highlights unusual patterns with contextual explanation

### Persona 2: Network Enthusiast / Lab User

**Background:** Individual operating a small home lab or test network
**Technical skill level:** Intermediate to advanced

**Goals:**
• Gain visibility into DNS activity on a local network
• Experiment with configuration changes and observe DNS behaviour

**Frustrations:**
• Raw DNS logs are difficult to interpret
• Existing dashboards provide limited behavioural context

**How the system helps:**
• Transforms raw logs into interpretable behavioural metrics
• Supports experimentation in a controlled environment

## 7.2 User Stories

The following user stories describe the core functionality of the system from the user's perspective.

• **US1:** As a user, I want to view DNS query volume over time so that I can understand normal traffic patterns.

• **US2:** As a user, I want to see the distribution of DNS response codes so that I can identify unusual error behaviour.

• **US3:** As a user, I want to view commonly queried domains and TLDs so that I can understand typical DNS usage.

• **US4:** As a user, I want unusual DNS behaviour to be highlighted so that I can investigate deviations from baseline patterns.

• **US5:** As a user, I want the system to explain why behaviour is highlighted rather than simply flagging it.

## 7.3 User Journey

A typical interaction with the system follows the sequence below:

1. The user deploys a DNS logging tool (Pi-hole or dnsmasq) within a controlled environment.
2. DNS logs are collected during normal system and application usage.
3. The user selects or uploads a DNS log dataset for analysis.
4. The analysis engine processes the logs and establishes baseline metrics.
5. The dashboard presents DNS behaviour trends, baseline comparisons, and highlighted observations.
6. The user interprets results and explores behavioural patterns.

At the midpoint stage, this user journey has been validated through the development of a proof-of-concept analysis pipeline.

## 7.4 Use Case Descriptions

### Use Case 1: Analyse DNS Logs

**Actor:**User
**Description:** The user submits DNS log data to the system for analysis.
**Outcome:** The system parses logs and computes behavioural metrics.

### Use Case 2: Visualise DNS Behaviour

**Actor:**User
**Description:** The user views dashboards displaying DNS behaviour trends and distributions.
**Outcome:** The user gains insight into normal DNS behaviour.

### Use Case 3: Identify Unusual Behaviour

**Actor:**User
**Description:** The user inspects highlighted deviations from baseline behaviour.
**Outcome:** The user understands what behaviour is unusual and why it was highlighted.

## 7.5 Functional Requirements

• The system shall parse DNS logs from Pi-hole and dnsmasq.
• The system shall compute baseline DNS behaviour metrics.
• The system shall highlight deviations from baseline behaviour using heuristic rules.
• The system shall visualise DNS behaviour through a web-based dashboard.

## 7.6 Non-Functional Requirements

• The system shall be lightweight and easy to deploy.
• The system shall prioritise interpretability over detection accuracy.
• The system shall operate only in controlled environments.
• The system shall comply with legal and ethical constraints.

# 8. Solution Design

## 8. 1High-Level System Architecture

The system is organised into four logical layers:

1. **Data Source Layer** – DNS logs generated by Pi-hole or dnsmasq within a controlled environment.
2. **Ingestion Layer** – Responsible for parsing raw log files and normalising records into a consistent internal format.
3. **Analysis Layer** – Computes behavioural features, baseline metrics, and heuristic observations.
4. **Presentation Layer** – Visualises metrics and highlighted observations through a web-based dashboard.

This layered architecture ensures that changes to log formats or visualisation components do not require modifications to the core analytical logic.

## 8.2 Data Flow and Processing Pipeline

DNS data flows through the system in a clear sequence of processing stages:

1. **Log acquisition:** DNS logs are collected from Pi-hole or dnsmasq.
2. **Parsing and normalisation:** Log entries are parsed and converted into structured records.
3. **Feature extraction:** Additional behavioural attributes are derived from each record.
4. **Baseline computation:** Normal DNS behaviour is characterised using aggregated metrics.
5. **Deviation analysis:** Heuristic rules identify deviations relative to established baselines.
6. **Visualisation:** Results are rendered in the dashboard for interpretation.

This pipeline-based design supports reproducibility and simplifies debugging by allowing each stage to be validated independently.

## 8.3 Data Model and Derived Features

### 8.3.1 Normalised DNS Record Schema

Each DNS query is represented internally using a normalised schema with the following core fields:

• Timestamp
• Queried domain name
• DNS response code
• Query type
• Log source (Pi-hole or dnsmasq)

Client identifiers are excluded or anonymised where applicable to maintain privacy.

### 8.3.2 Derived Behavioural Features

From the normalised records, the system derives additional behavioural features, including:

• Top-level domain (TLD)
• Domain name length
• Subdomain depth (label count)
• Time-bucketed query counts (e.g., per minute or per hour)

These derived features enable analysis of DNS structure, temporal patterns, and distributional characteristics without requiring packet inspection or payload analysis.

## 8.4 Core Modules and Responsibilities

The system design is decomposed into the following core modules:

**LogParser:**
Detects log format (Pi-hole or dnsmasq), extracts relevant fields, and safely handles malformed entries.

**FeatureExtractor:**
Computes derived attributes such as TLD, domain length, and label depth, and aggregates records into time buckets.

**MetricsEngine:**
Calculates baseline metrics including query volume trends, response code distributions, and domain popularity.

**HeuristicEngine:**
Applies transparent rule-based heuristics to identify potentially unusual DNS behaviour and produces explanatory observations.

**DashboardController:**
Loads analytical outputs and renders charts, tables, and summaries for user interpretation.

This modular design supports independent testing and future extensibility.

## 8.5 Design Artefacts

Industry-standard design artefacts are used to document the system structure and interactions.

•UML class diagrams describe relationships between analysis components.
•UML sequence diagrams illustrate key use cases such as DNS log analysis and dashboard rendering.
•Data flow diagrams capture the end-to-end processing pipeline.

Due to space constraints, these diagrams are included in the appendix and referenced where relevant. The appendix will contain all figures referenced in this section, including UML class diagrams, sequence diagrams, and data flow diagrams.

## 8.6 Design Constraints and Assumptions

The system design is based on the following constraints and assumptions:

• All DNS data is generated within a controlled environment.
• Data volumes are modest and suitable for local processing.
• Behavioural interpretation prioritises clarity over exhaustive detection.
• The system is not intended for real-time enterprise deployment.

These constraints ensure alignment with the project scope and ethical considerations.

# 9. Solution Implementation

This section describes the implementation work completed up to the midpoint stage of the project and focuses on the development of a functional proof-of-concept (PoC). The purpose of the PoC is to validate the feasibility of the proposed system architecture, confirm the suitability of selected tools, and reduce technical risk at an early stage. At this point in the project, the implementation prioritises end-to-end data processing using limited and controlled DNS datasets rather than full feature completeness or large-scale evaluation.

Implementation has been carried out incrementally, with early emphasis placed on DNS log ingestion, parsing, and behavioural feature extraction. This staged approach ensures that core analytical components are functioning correctly before further expansion of heuristics, datasets, and visualisation capabilities in the final phase of the project.

## 9.1 Implementation Scope at Midpoint

At midpoint stage, the following components have been implemented and validated through the proof-of-concept:

- DNS log ingestion from lightweight DNS log sources
- Parsing and normalisation of DNS log entries
- Extraction of core behavioural features
- Computation of basic DNS behaviour metrics
- Execution of a subset of heuristic rules to highlight unusual behaviour

Components such as full dashboard visualisation, extensive heuristic tuning, and large-scale dataset evaluation are planned for later stages and are not within the scope of the current PoC.

## 9.2 DNS Log Ingestion and Parsing

The current implementation supports DNS logs generated by lightweight tools such as Pi-hole and dnsmasq within a controlled local environment. DNS logs are exported in text-based formats and provided as input to the analysis engine.

A Python-based log parsing module has been implemented to process raw DNS logs line-by-line. The parser detects the log source format and extracts key fields required for analysis, including timestamps, queried domain names, and DNS response codes. To ensure robustness, malformed or incomplete log entries are safely ignored rather than causing processing failures.

The parsing process converts heterogeneous log formats into a consistent internal representation, enabling downstream analysis to operate independently of the original log source.

## 9.3 Feature Extraction and Behavioural Metrics

Following parsing, a feature extraction module derives additional behavioural attributes from each DNS record. At midpoint stage, the following features are implemented:

- Extraction of top-level domains (TLDs)
- Calculation of domain name length
- Identification of subdomain depth (label count)
- Aggregation of DNS query counts

Using these features, basic DNS behaviour metrics are computed, including total query volume, response code counts, domain popularity, and TLD distribution. These metrics provide an initial baseline for understanding DNS behaviour within the controlled environment.

The proof-of-concept demonstrates that meaningful behavioural features can be derived using DNS metadata only, without requiring packet inspection or payload data.

## 9.4 Heuristic Identification of Unusual Behaviour

A heuristic analysis module has been implemented to support identification of DNS behaviour that deviates from observed baseline patterns. Heuristics are defined as transparent, rule-based checks applied to extracted behavioural features rather than as security detection mechanisms.

At midpoint stage, the long-domain heuristic has been fully exercised using sample DNS logs. This heuristic identifies domain names whose length exceeds typical values observed within the dataset and highlights them as potentially unusual behaviour. For example, during PoC testing, a deeply nested domain containing multiple subdomain labels was correctly identified and reported as unusual.

Additional heuristics, including repeated NXDOMAIN responses, sudden bursts in DNS query volume, and rare TLD usage, are implemented at rule-definition level. These heuristics are designed and documented but will be validated using larger and more varied datasets during the final phase of the project.

Each heuristic produces an explanatory observation describing why the behaviour was highlighted, ensuring that results are interpretable and aligned with the project’s exploratory scope.

# 9.5 Proof-of-Concept Validation

The proof-of-concept was validated using a small, controlled DNS log dataset generated during normal system usage. A deliberately limited dataset was used at this stage to allow clear inspection of parsing logic, feature extraction, and heuristic behaviour before introducing larger and noisier datasets.

The PoC successfully demonstrates that:

- Lightweight DNS logs are sufficient for behavioural analysis
- DNS metadata can be reliably parsed and normalised
- Behavioural features and metrics can be extracted consistently
- Heuristic rules can highlight unusual DNS behaviour without assigning malicious intent

This early validation confirms the feasibility of the proposed system architecture and significantly reduces technical risk.

# 9.6 Current Limitations and Planned Work

## 9.6.1 Current Limitations

At midpoint stage, the implementation has the following limitations:

- Use of a small DNS dataset intended for feasibility validation rather than comprehensive analysis
- Limited heuristic execution, with only a subset fully exercised
- Dashboard visualisation remaining at prototype or planning stage

These limitations are expected and appropriate at this stage of the project.

## 9.6.2 Planned Implementation Work

Planned work for the final phase of the project includes:

- Expansion of heuristic validation using larger and more varied DNS datasets
- Refinement of heuristic thresholds based on observed baseline behaviour
- Integration of the analysis engine with an interactive dashboard
- Extended evaluation of interpretability and usability

# 10. Solution Testing / Evaluation

## 10.1 Evaluation Objectives

The evaluation is designed to address the following questions:

• Can the system reliably parse DNS logs from lightweight sources such as Pi-hole and dnsmasq?
• Are baseline DNS behaviour metrics stable and meaningful across datasets?
• Do heuristic rules highlight deviations in a consistent and explainable way?
• Can users interpret the visualised results without specialist DNS expertise?
• What limitations arise from the chosen analytical approach?

These objectives are directly aligned with the project aims and scope.

## 10.2 Evaluation Methodology

Evaluation will be conducted using a combination of technical testing, dataset-based behavioural analysis, and user-oriented usability evaluation.

### 10.2.1 Technical Evaluation

Technical evaluation will verify the correctness and robustness of the analysis engine.

Planned technical tests include:

•Validation of DNS log parsing accuracy
•Verification of baseline metric computation
•Consistency of outputs across repeated executions
• Error handling for malformed or incomplete log entries

### 10.2.2 Dataset-Based Behavioural Evaluation

Behavioural evaluation will assess how effectively the system characterises DNS behaviour and highlights deviations using multiple controlled datasets.

Planned datasets include:

•**Dataset A:** Baseline DNS traffic generated during normal daily system usage
•**Dataset B:** Burst traffic produced during system startup, application launches, and updates
•**Dataset C:** Controlled repeated queries resulting in NXDOMAIN responses
•**Dataset D:** Queries containing long domain names and deep subdomain structures

Evaluation will focus on baseline stability, sensitivity of heuristics to controlled deviations, and avoidance of excessive false positives during normal activity.

### 10.2.3 Dashboard Usability Evaluation

As the system includes a user-facing dashboard, a lightweight usability evaluation will be conducted.

Planned approach:

•Think-aloud usability sessions with representative users (e.g., cyber security students)
•Participants complete interpretation tasks without prior explanation of metrics

Example tasks include:

•Identifying periods of highest DNS query activity
•Interpreting response code distributions
• Explaining why specific behaviour was highlighted as unusual

Feedback will be collected on clarity, usability, and interpretability.

## 10.3 Evaluation Metrics

The following metrics will be used to assess system performance:

| Evaluation Area | Metric | Measurement Method | Success Criteria |
|---|---|---|---|
| Log Parsing | Parsing success rate | Percentage of valid entries processed | ≥ 95% |
| Baseline Stability | Metric consistency | Repeated runs on same dataset | Minimal variation |
| Behaviour Sensitivity | Deviation detection | Known deviations flagged | Deviations identified |
| False Positives | Benign flag rate | Flags during baseline traffic | Low frequency |
| Interpretability | User task success | Think-aloud completion | Majority successful |

## 10.4 Limitations of the Evaluation

The planned evaluation has the following limitations:

• Datasets are limited to controlled environments
• Results may not generalise to enterprise-scale networks
• Heuristic rules may occasionally flag benign behaviour

These limitations are acknowledged and will inform recommendations for future work.

# Supervisor Meeting Records

**Supervisor:** Badis Aoun
**Project Title:** DNS Behaviour Monitor

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## Meeting 1: Project Selection & Initial Guidance

- Project topic discussed and confirmed as DNS behaviour analysis
- Supervisor explained assessment structure and marking criteria
- Guidance provided on project proposal poster requirements
- Emphasis placed on appropriate scope for a BSc Cyber Security project

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## Meeting 2: Poster Review & Formative Feedback

- Poster reviewed and feedback provided
- Evaluation recommended to focus on **user understanding of DNS behaviour**, not only technical correctness
- Use of **Kanban board** suggested to manage backlog, milestones, and progress tracking
- Literature review advised to be **more summarised and analytical**, avoiding excessive description
- Project aims recommended to be clarified and strengthened
- Feedback encouraged to be treated as **user stories** representing fellow students
- Importance of keeping poster feedback and meeting records highlighted

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## Meeting 3: Midterm Review

- Supervisor advised listing, analysing, and summarising key academic papers
- Competitor DNS tools to be explicitly listed and compared
- Recommendation to develop a clear **legal, ethical, and professional considerations** section

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## Reflection

- Supervisor feedback directly informed changes to literature review structure, evaluation approach, project management, and ethical considerations
- Regular meetings ensured the project remains realistic, ethically compliant, and aligned with assessment criteria

# 11. References

[1] P. V. Mockapetris, “Domain Names – Concepts and Facilities,” RFC 1034, Nov. 1987.

[2] P. V. Mockapetris, “Domain Names – Implementation and Specification,” RFC 1035, Nov. 1987.

[3] J. Jung, E. Sit, H. Balakrishnan, and R. Morris, “DNS Performance and the Effectiveness of Caching,” IEEE/ACM Transactions on Networking, vol. 10, no. 5, pp. 589–603, Oct. 2002.

[4] M. Crovella and B. Krishnamurthy, Internet Measurement: Infrastructure, Traffic and Applications. Hoboken, NJ, USA: Wiley, 2006.

[5] DNS Operations, Analysis, and Research Center (DNS-OARC), “DNS Behaviour Analysis Reports,” 2023. [Online]. Available: https://www.dns-oarc.net. [Accessed: Jan. 2026].

[6] Cloudflare Research, “Understanding DNS Traffic Patterns,” 2023. [Online]. Available: https://www.cloudflare.com. [Accessed: Jan. 2026].

[7] Pi-hole Developers, “Pi-hole Documentation,” 2024. [Online]. Available: https://docs.pi-hole.net. [Accessed: Jan. 2026].

# 12. Appendices

Appendices provide supporting material that complements the main body of the report. The appendices are referenced where relevant throughout the report but are not required to understand the core project narrative.

## Appendix A – Project Proposal Poster

### Problem Statement

DNS is a core internet service, but its behaviour is rarely examined in everyday networks. Normal DNS traffic follows predictable patterns, while unusual behaviour—such as long domain names, repeated requests, or uncommon TLDs may indicate misconfigurations, automation, or unexpected network activity.

This project aims to analyse DNS behaviour by studying DNS logs and creating a lightweight monitoring tool that visualises patterns and highlights unusual DNS behaviours. The goal is to better understand DNS trends, not to build a security or detection product.

### Aims and Objectives

Aims

- Analyse DNS behaviour and characterise normal patterns.
- Identify unusual or unexpected DNS behaviours.
- Develop a lightweight DNS behaviour monitoring tool with visual outputs.

Objectives

- Collect DNS logs using Pi-hole or dnsmasq.
- Identify baseline DNS patterns (frequency, TLDs, response codes).
- Detect unusual behaviours (long domains, repeated NXDOMAIN, rare TLDs).
- Build a Python behaviour-analysis engine.
- Create a dashboard for visualisation (Flask/Streamlit).
- Highlight unusual behaviours (non-security focused).
- Test with varied DNS traffic datasets.
- Evaluate results and document findings.

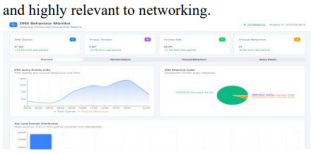
### DNS Behaviors Monitor: Analyzing Normal and Unusual DNS Patterns in Network DNS Traffic

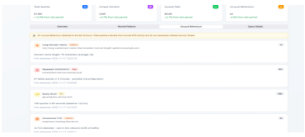
By Sajan Tamang

### Introduction

DNS enables devices to access online services by translating domain names into IP addresses. Although DNS queries occur continuously, their behaviour is rarely analysed beyond basic functionality. Understanding DNS patterns can reveal how devices interact with services, what domains are commonly used, and how activity changes over time.

This project examines DNS behaviour by analysing logs from a controlled environment. It studies typical patterns such as domain popularity, query frequency, time intervals, TLD distributions, and response codes. It also identifies unusual behaviour such as long domain structures, repeated failed lookups, uncommon TLDs, or sudden query bursts.A lightweight DNS behaviour monitoring tool will be developed to process logs, visualise data trends, and highlight unusual patterns. The focus is on understanding DNS behaviour itself rather than detecting threats, making the project exploratory, analytical, and highly relevant to networking.





### Key Milestones

Nov 17 – Nov 30: Research & setup
Dec 1 – Dec 20: Log parsing + behaviour analysis
Dec 21 – Jan 5: Light work during break
Jan 6 – Jan 12: Mid-Review prep
Jan 12 – Jan 26: Mid-Project Review
Jan 27 – Feb 20: Dashboard development
Feb 21 – Mar 10: DNS behaviours highlighting
Mar 11 – Apr 5: Testing & evaluation
Apr 6 – Apr 30: Artefact finalisation
May 2026: Artefact demo
1–20 May 2026: Final report & screencast submission

### Legal, Ethical & Professional

This project analyses DNS metadata only—such as domain names, timestamps, and DNS response codes. No personal content or sensitive user information is collected. All DNS logs are generated within a controlled local environment and are anonymised where necessary. Logs will be securely stored and deleted after the project, ensuring full compliance with GDPR.

The project complies with the UK Computer Misuse Act (1990) as no unauthorised access, network scanning, or external monitoring is performed. All tools used (Pi-hole, dnsmasq, analysis libraries) are open source and properly referenced.

Following BCS ethical and professional standards, the project prioritises data minimisation, user privacy, transparency, and responsible research conduct. Its purpose is analytical and educational, not investigative or intrusive.

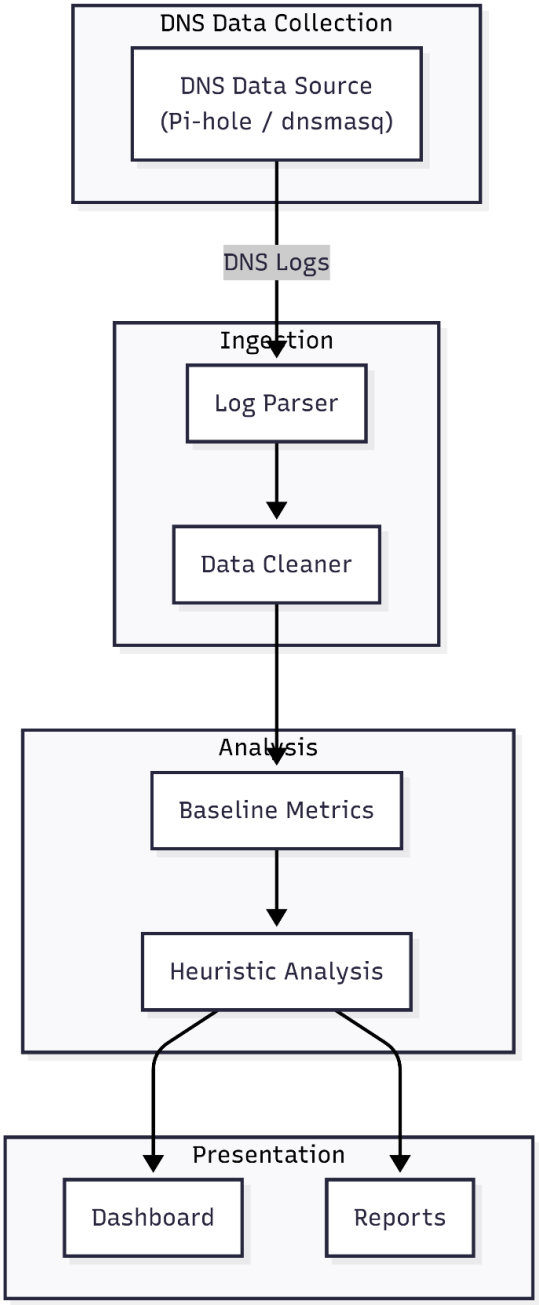
### Key References

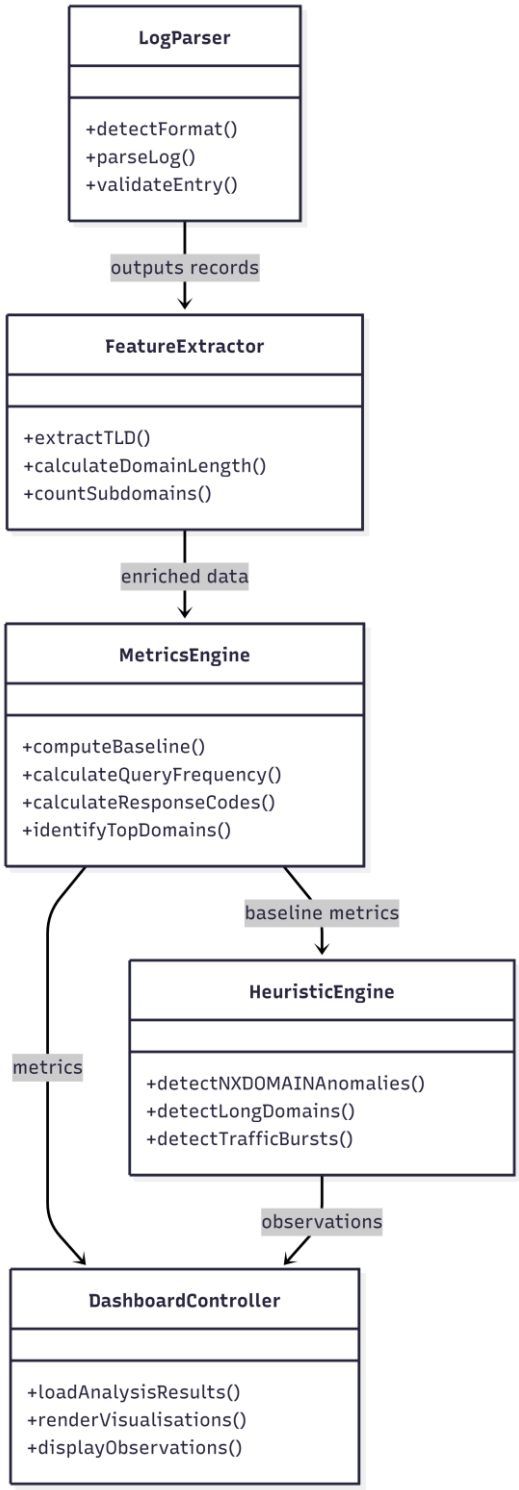
[1] W. Stallings, Network Security Essentials: Applications and Standards, 7th ed., Pearson, 2020.
[2] DNS-OARC, "DNS Behaviour Analysis Reports," 2023.
[3] Cloudflare, "Understanding DNS Traffic Patterns," Cloudflare Research, 2023.
[4] S. Hathaway, "Statistical Analysis of DNS Traffic," IEEE CNS, pp. 1–8, 2022.
[5] Pi-hole Developers. Pi-hole Documentation, 2024.

### Use of AI

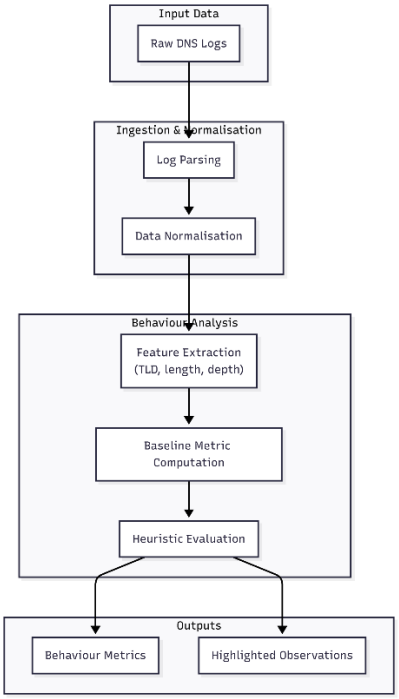
AI tools such as ChatGPT were used to refine written sections, summarise background research, and support planning. All coding, analysis logic, visualisations, and interpretation will be created manually. AI-generated text is cross-checked with authoritative technical sources to avoid inaccuracies or bias. Any AI assistance will be transparently declared in the final project report in accordance with university academic integrity guidelines

## Appendix B – System Architecture and UML Diagrams

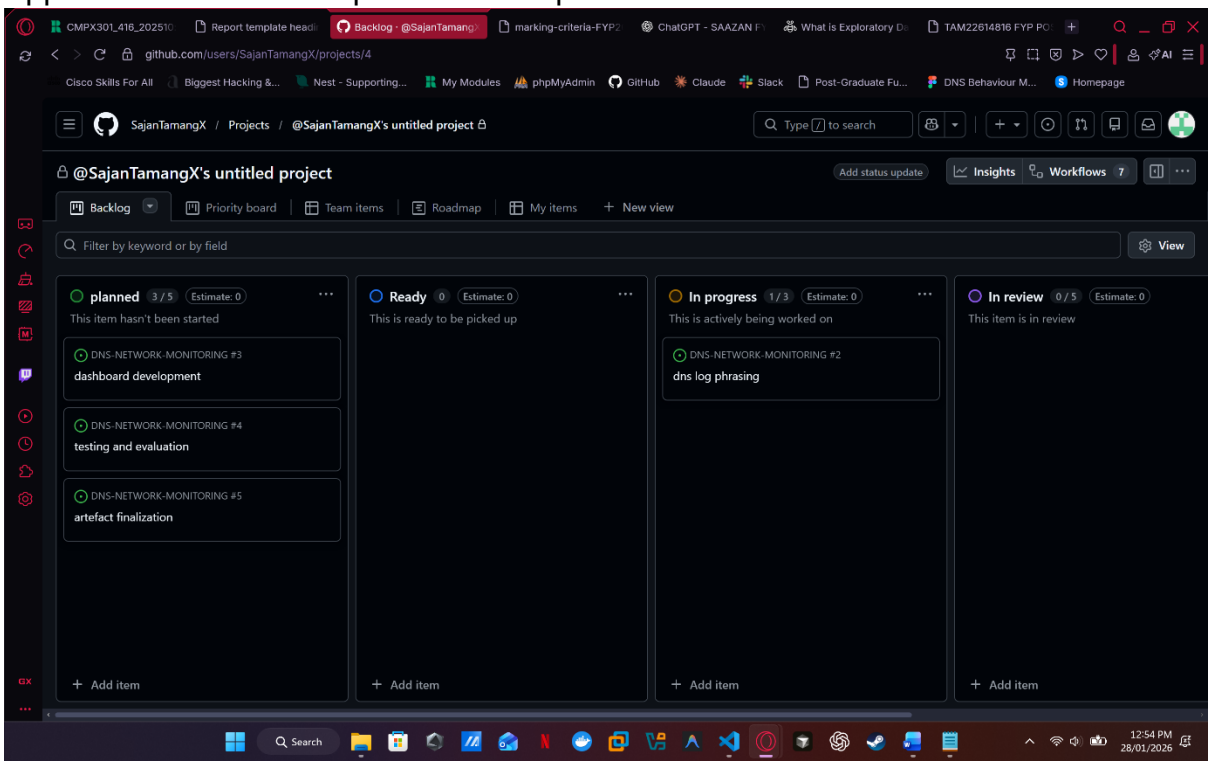


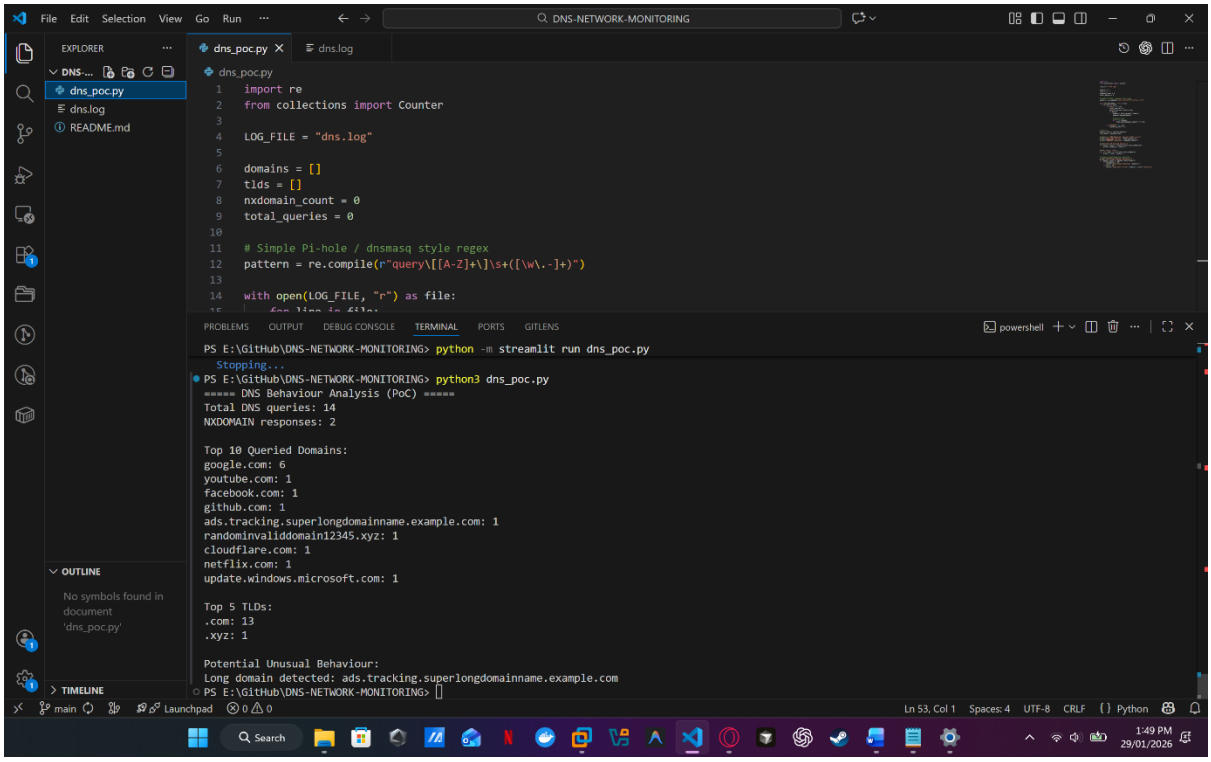




## Appendix C – Data Flow Diagrams




## Appendix D: Github and proof of concept




File Edit Selection View Go Run
DNS-NETWORK-MONITORING
EXPLORER
DNS-...
dns_poc.py
dns.log
README.md
OUTLINE
No symbols found in document 'dns_poc.py'
TIMELINE
dns_poc.py
dns.log
import re
from collections import Counter
LOG_FILE = "dns.log"
domains = []
tlds = []
nxdomain_count = 0
total_queries = 0
# Simple Pi-hole / dnsmasq style regex
pattern = re.compile(r"query\[[A-Z]+\]\s+([\w\.-]+)")
with open(LOG_FILE, "r") as file:
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
powershell
PS E:\GitHub\DNS-NETWORK-MONITORING> python -m streamlit run dns_poc.py
Stopping...
PS E:\GitHub\DNS-NETWORK-MONITORING> python3 dns_poc.py
===== DNS Behaviour Analysis (PoC) =====
Total DNS queries: 14
NXDOMAIN responses: 2
Top 10 Queried Domains:
google.com: 6
youtube.com: 1
facebook.com: 1
github.com: 1
ads.tracking.superlongdomainname.example.com: 1
randominvaliddomain12345.xyz: 1
cloudflare.com: 1
netflix.com: 1
update.windows.microsoft.com: 1
Top 5 TLDs:
.com: 13
.xyz: 1
Potential Unusual Behaviour:
Long domain detected: ads.tracking.superlongdomainname.example.com
PS E:\GitHub\DNS-NETWORK-MONITORING>
main
Launchpad
Ln 53, Col 1
Spaces: 4
UTF-8
CRLF
Python
Search
1:49 PM
29/01/2026